SPAN 2

The Complete Implementation Guidebook

Every Output, Every Formula, Every Decision Explained

8 Modules · Real Market Data · Historical VaR · Stress VaR · Vol Surface · Options

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Part I — The Big Picture

What Is SPAN 2 and Why Does It Exist?

SPAN 2 (Standard Portfolio Analysis of Risk, version 2) is CME Group's next-generation margin methodology, launched in 2022 to replace the original SPAN system that had been running since 1988. To understand why SPAN 2 was needed, we first need to understand the weakness of the original SPAN.

The Problem with Classic SPAN

Classic SPAN calculates margin by testing your portfolio against 16 fixed, hand-crafted scenarios — for example, 'price moves up by the full scan range' or 'price moves down by two-thirds of the scan range'. These 16 scenarios are chosen by exchange analysts and updated periodically. The weakness is threefold:

1. The 16 scenarios may miss the actual worst-case event. The COVID oil crash of April 2020 — when crude oil went negative — was not one of the 16 scenarios anyone had pre-defined.
2. Scan ranges are updated slowly. In a fast-moving market, the scan range may be weeks behind actual volatility.
3. Options are priced using flat, hardcoded implied volatility numbers that do not reflect the current vol surface, undercharging margin during high-vol regimes.

The SPAN 2 Solution

SPAN 2 replaces the fixed 16-scenario grid with a Historical Value at Risk (HVaR) framework. Instead of asking 'what if prices move by our assumed scan range?', it asks: 'what actually happened to prices over the last 10 years — and what would our portfolio have lost on the worst 1% of those days?'

> Core Insight: If you replay 10 years of actual daily market moves (about 2,500 trading days) against your portfolio, the worst 1% of days — the worst 25 days — gives you a statistically grounded measure of your real tail risk. This is what HVaR means.
>
> SPAN 2 then adds: a stress component (SVaR) anchored to the worst historical crisis periods, a vol surface for options, and liquidity/concentration charges for large positions.

The SPAN 2 Margin Formula

$$\text{SPAN 2 Margin} = (x \times \text{HVaR}) + ((1-x) \times \text{SVaR}) + \text{Liquidity} + \text{Concentration}$$

Where x = 40% (HVaR weight), (1-x) = 60% (SVaR weight) in our model

Breaking this down:

Component	Symbol	Weight	What It Captures

Historical VaR	HVaR	40%	Worst 1% of last 10 years of vol-scaled daily returns
Stress VaR	SVaR	60%	Worst loss during specifically defined crisis periods (COVID, GFC, etc.)
Liquidity Charge	Liq	Additive	Cost of unwinding position at bid-ask spread
Concentration Charge	Conc	Additive	Extra margin if position is large vs market depth (ADV)

Why 40% HVaR and 60% SVaR?

The higher weight on SVaR is deliberate and addresses a phenomenon called pro-cyclicality. In calm markets, HVaR naturally falls because recent history has few large losses — this would cause margin requirements to drop exactly when complacency is highest. By keeping 60% anchored to the worst historical stress periods, SPAN 2 ensures margin never collapses to dangerously low levels during quiet times.

> Analogy: Imagine insuring your car. HVaR is like setting your premium based on your recent driving — if you have had no accidents in 3 years, your premium would drop. SVaR is like saying your premium also depends on the worst-ever claims from your risk category regardless of your recent record. The blend prevents you from becoming under-insured during good times.

Part II — Complete Glossary

Every Term, Abbreviation and Symbol Used in This Model

This glossary is comprehensive. Every single abbreviation that appears in the model output is defined here. Read this once before studying the module outputs.

A. Core Statistical & Risk Terms

Term / Symbol	Full Form	Plain English Explanation	Where You See It
HVaR	Historical Value at Risk	The loss that is exceeded only 1% of the time based on replaying actual historical market moves. With 2,701 scenarios, the 99th percentile = worst 27 days.	Module S3 output: HVaR (99%)
SVaR	Stress Value at Risk	HVaR computed specifically on a pre-defined crisis period (e.g., COVID). Captures risk that may be underweighted in the full 10yr window.	Module S4: SVaR by stress period
ES	Expected Shortfall (also called CVaR)	The average loss across ALL scenarios in the tail — not just the worst one. More stable than VaR because it averages the tail rather than picking a single point.	Module S3: Expected Shortfall
CVaR	Conditional Value at Risk	Same as Expected Shortfall. The 'conditional' refers to conditioning on being in the tail.	Synonymous with ES
VaR	Value at Risk	General term for the loss at a given confidence level. HVaR and SVaR are both types of VaR.	Throughout
99% Confidence	—	We set the margin at a level that covers losses in 99% of scenarios. Only the worst 1% of days would exceed the margin.	hvar_confidence = 0.99
Tail	—	The extreme end of the loss distribution — the worst 1% of scenarios (27 out of 2,701 in our model).	n_tail = 27
Fat Tails	—	When extreme events happen more often than a normal distribution predicts. Excess kurtosis > 0 indicates fat tails.	Kurtosis: CL=22, NG=25, GC=10

Skewness	—	Asymmetry of the distribution. Negative skew = large crashes more likely than large rallies. Positive skew = large rallies more likely.	NG skew = -1.005 (crash risk)
Kurtosis	—	Measure of tail fatness. Normal distribution = 0. Higher kurtosis = more extreme events occur than expected.	CL excess kurtosis = 22.262

B. SPAN 2 Specific Terms

Term	Full Form / Meaning	Explanation
Vol Scaling	Volatility Scaling	Rescaling historical returns to today's vol level. A 2015 return that was 'normal' then becomes appropriately larger when replayed at today's higher vol.
Scaled Return	—	r_scaled = r_raw × (current_vol / vol_at_t). Historical returns rescaled to current volatility environment.
Rolling Vol	Rolling Volatility	60-day window of daily returns, annualised by multiplying by √252. Captures how volatile the market currently is.
Blended Risk	—	The combination: x×HVaR + (1-x)×SVaR. The primary risk number before adding liquidity/concentration charges.
Floor	80% Floor	Blended risk ≥ 80% × max(HVaR, SVaR). Prevents the blend from being too low when one component dominates.
Pro-cyclicality	—	When margin falls during calm markets and rises during crashes — exactly backwards from what is needed. SVaR weight prevents this.
Binding Period	—	The stress period that produces the highest SVaR. In our model, COVID is binding at $80,319.
ADV	Average Daily Volume	The average number of lots traded per day in a contract. Used to detect concentration.
Concentration	—	When a position is large relative to ADV — meaning liquidating it would move the market against you.
Liquidity Horizon	—	The assumed number of days needed to fully unwind a position. 1 day for liquid futures.
Attribution	—	How much each position contributes to portfolio HVaR. Computed as average P&L of each position across the 27 worst portfolio days.

Diversification Benefit	—	The reduction in combined HVaR vs the sum of standalone HVaRs. Arises because positions do not all lose money on the same days.

C. Options Terms

Symbol	Name	Formula / Definition	In Our Model
F	Futures Price	Current settlement price of the underlying futures contract	CL: $79.20, GC: $2,025
K	Strike Price	The fixed price at which the option can be exercised	CL call: $82.00, CL put: $76.00
T	Time to Expiry	Years until the option expires. 30 days = 30/365 = 0.0822yr	CL options: 30/365
σ (sigma)	Implied Volatility (IV)	Market's annualised forecast of price variability	CL flat IV: 28%, Surface IV: 54%
r	Risk-Free Rate	Interest rate (set to 0 for margining purposes)	r = 0.0
N(x)	Cumulative Normal Distribution	Probability that a standard normal variable is ≤ x	Used in Black-76 formula
d1	Black-76 d1	$[\ln(F/K) + 0.5\sigma^2T] / (\sigma\sqrt{T})$	Intermediate calculation
d2	Black-76 d2	$d1 - \sigma\sqrt{T}$	Intermediate calculation
Δ (Delta)	Delta	Rate of change of option price with futures price. Call delta ∈ (0,1), Put delta ∈ (-1,0)	CL C82: 0.4420, CL P76: -0.3666
Γ (Gamma)	Gamma	Rate of change of delta. High gamma = delta changes rapidly as price moves	Measures curvature of option P&L
v (Vega)	Vega	Option price change per 1% move in implied vol (per 0.01 change in σ)	GC C2100: 2.7751 per vol point
Θ (Theta)	Theta	Option price loss per calendar day (time decay). Always negative for long options	Not displayed but embedded in pricing
OTM	Out of the Money	Call: F < K (futures below strike). Put: F > K (futures above strike)	All 3 options in our portfolio are OTM
ITM	In the Money	Call: F > K. Put: F < K. Exercising would be profitable now	Not in our portfolio
ATM	At the Money	F ≈ K. Delta ≈ 0.5 for calls, -0.5 for puts	Reference point for vol surface
IV	Implied Volatility	The vol that makes Black-76 price equal the observed market price	Flat IV vs Surface IV differs by ~2,600bp

Vol Surface	Implied Vol Surface	A grid of IVs across strike (moneyness) and expiry (tenor). Captures the smile and skew.	Built in Module S5
Smile	Vol Smile	The U-shape of IV across strikes — OTM options are more expensive than ATM	CL: negative skew, NG: positive skew
Skew	Vol Skew	Asymmetry of the smile. Negative skew = puts more expensive than calls (crude oil typical)	CL skew b = -0.08
Moneyness	—	K/F — ratio of strike to futures price. 1.0 = ATM, <1.0 = OTM put, >1.0 = OTM call	Surface grid: 80% to 120%
SOM	Short Option Minimum	Floor margin for short options. Prevents near-zero margin for deep OTM shorts.	Not binding in our model (scan > SOM)
Black-76	—	Options pricing model for futures options. Variant of Black-Scholes for futures.	Used in Module S5, S6

D. Market & Product Terms

Abbreviation	Full Form	Meaning
CME	Chicago Mercantile Exchange	World's largest futures exchange, creator of SPAN/SPAN 2
CL	WTI Crude Oil	West Texas Intermediate crude oil futures. $1/bbl move = $1,000/lot (1,000 bbl/contract)
NG	Natural Gas	Henry Hub natural gas futures. $1/MMBtu move = $10,000/lot (10,000 MMBtu/contract)
GC	Gold	COMEX gold futures. $1/oz move = $100/lot (100 oz/contract)
WTI	West Texas Intermediate	The US benchmark crude oil grade, priced at Cushing, Oklahoma
MMBtu	Million British Thermal Units	Unit of natural gas energy. Standard contract = 10,000 MMBtu
ADV	Average Daily Volume	Average lots traded per day. CL: 350,000 lots/day, NG: 120,000, GC: 180,000
P&L	Profit and Loss	Money gained or lost on a position
bbl	Barrel	Unit of crude oil. 1 barrel = 42 US gallons
bp	Basis Points	1 bp = 0.01%. Used to express small differences in implied volatility
yfinance	Yahoo Finance Python library	Python package for downloading free historical market data

NYMEX	New York Mercantile Exchange	CME subsidiary that lists CL and NG futures
COMEX	Commodity Exchange	CME subsidiary that lists GC (gold) futures

Part III — Module S1: Data Layer

What This Module Does

Module S1 is the foundation — it defines every input the model needs. Nothing is calculated here; this module is purely about specifying what contracts exist, what positions are held, and what parameters govern the margin calculation.

S1.1 Contract Specifications (ContractSpec)

Each futures contract has fixed exchange-defined properties. Our model tracks seven fields per contract:

Field	Type	Example (CL_Mar25)	Why It Matters
product_code	String	CL	Groups contracts by product for spread detection
expiry	String	Mar-25	Identifies which delivery month — needed for calendar spread pairing
point_value	Dollar	$1,000	Converts price moves to dollar P&L. CL moves $1 → each lot gains/loses $1,000
tick_size	Dollar	$0.01	Minimum price increment. Not used in HVaR but needed for real-world pricing
yf_ticker	String	CL=F	The Yahoo Finance ticker used to download 10yr price history in Module S2
adv_lots	Lots	350,000	Average daily volume — used in Module S7 for concentration charge threshold
liquidity_horizon_days	Days	1	Assumed days to unwind — used in liquidity charge. Liquid futures = 1 day

S1.2 Portfolio Positions

Contract	Direction	Lots	Settlement	Why Included
CL_Mar25	LONG	+5	$79.20	Part of CL calendar spread. Long leg, 5 lots
CL_Jun25	SHORT	-3	$80.100	Short leg of CL calendar spread, 3 lots. Partially offsets Mar25
NG_Mar25	LONG	+4	$2.710	Natural gas long — correlated with CL for inter-commodity spread

GC_Apr25	LONG	+2	$2,025.00	Gold long — different asset class, minimal correlation with energy

S1.3 SPAN 2 Global Parameters (SPAN2GlobalParams)

These parameters control the core SPAN 2 methodology. Understanding them is critical:

Parameter	Value	Full Name	Explanation
hvar_confidence	0.99	HVaR Confidence Level	We require margin to cover 99% of historical scenarios. Only 1% of days (the worst 27 out of 2,701) are allowed to exceed the margin.
hvar_lookback_years	10	HVaR Lookback Window	We use 10 years of daily price history. CME uses at least 10 years as a regulatory requirement.
hvar_weight (x)	0.40	HVaR Blending Weight	40% of final margin comes from HVaR. Reflects current-market-calibrated risk.
svar_weight (1-x)	0.60	SVaR Blending Weight	60% of final margin comes from SVaR. Ensures stress periods always dominate margin.
liq_charge_multiplier	0.50	Liquidity Charge Multiplier	We charge 50% of the full bid-ask spread cost — assuming half the cost is unavoidable when unwinding.
conc_charge_rate	0.25	Concentration Charge Rate	25% additional margin on the portion of a position that exceeds the concentration threshold.

S1.4 Stress Periods

Four historical crisis windows are defined. Module S4 extracts P&L from these specific windows to compute SVaR.

Label	Date Range	Trading Days in Data	Key Event
GFC	Sep 2008 – Mar 2009	0 (data starts 2015)	Global Financial Crisis — commodity selloff. Not in our dataset.
OIL_CRASH_2014	Jun 2014 – Jan 2016	157	OPEC supply glut. WTI fell from $107 to $26/bbl. NG correlated.
COVID	Feb 2020 – May 2020	80	COVID crash + WTI went NEGATIVE on Apr 20, 2020. BINDING period.

ENERGY_SPIKE_2022	Feb 2022 – Aug 2022	147	Russia-Ukraine war. NG tripled, WTI hit $130. Gold safe haven.

Part IV — Module S2: Market Data Engine

What This Module Does

Module S2 downloads 10+ years of daily futures price history from Yahoo Finance and transforms it into a scenario set — a table of daily dollar price moves for each product, ready to be applied to the portfolio.

S2.1 Price Download

The module downloads daily closing prices for CL=F, NG=F, and GC=F. After removing dates where any product has missing data (dropna), we have 2,763 aligned trading days from March 2015 to March 2026.

> Why front-month futures? Yahoo Finance provides front-month (nearest expiry) futures prices for CL=F, NG=F, GC=F. These roll automatically as the nearest contract expires. This is imperfect — real SPAN 2 uses a full futures curve — but it gives us the correct price dynamics and volatility characteristics for a learning model.

S2.2 Log Returns

Daily log returns are computed as:

```
r(t) = ln[Price(t) / Price(t-1)]
```

Log returns are preferred over simple returns because they are additive across time and better-behaved statistically for fat-tailed distributions

S2.3 Volatility Scaling — The Key SPAN 2 Innovation

Raw historical returns from 2015 (when CL vol was ~25%) would understate today's risk (CL vol = 56%). SPAN 2 rescales every historical return to today's vol level:

```
r_scaled(t) = r_raw(t) × [current_vol / vol_at_t]
```

If a 2015 daily move was 'normal' at 25% vol, it becomes proportionally larger when scaled to today's 56% vol environment

Where:

Term	Calculation	Our Values
current_vol	60-day rolling annualised vol on the most recent day	CL: 56.4%, NG: 185.2%, GC: 39.2%
vol_at_t	60-day rolling annualised vol centred on date t	Varies by date — high in 2020/2022, low in 2016-2019
Rolling vol window	std(daily returns) × √252	252 = trading days per year, converts daily vol to annual

S2.4 Scenario Set Construction

For each historical date t and product p, the dollar price move is:

```
dollar_move(t, p) = current_price(p) × [exp(r_scaled(t, p)) - 1]
```

This converts the dimensionless scaled log return into an actual dollar move applicable to today's portfolio

The result is a 2,701 × 3 matrix. Each row is one scenario (one historical trading day), each column is one product (CL, NG, GC). The value in each cell tells us: 'if today's market replicated what happened on that historical date, how many dollars would the CL price move?'

S2.5 What the Return Statistics Tell You

Statistic	CL (Crude Oil)	NG (Natural Gas)	GC (Gold)	Interpretation
Ann. Volatility	45.3%	67.7%	16.1%	NG is 4x more volatile than GC. This drives NG's large contribution to HVaR.
Skewness	0.076	-1.005	-0.711	NG and GC have negative skew — large crashes more likely than large rallies. CL is roughly symmetric.
Excess Kurtosis	22.262	24.934	10.545	Fat tails across all products. Normal distribution = 0. These values mean extreme daily moves are ~22x more common than a normal dist predicts for CL.
Min Daily Return	-28.22%	-64.40%	-12.07%	NG's worst single day was a 64% crash — this is why NG is the dominant risk driver.
Worst 1% Return	-7.36%	-10.78%	-2.69%	At 99% confidence, CL can lose 7.36% in a single day — applied to 5 long lots × $79.20 × $1,000 = ~$29,000 loss.

Part V — Module S3: Historical VaR Engine

What This Module Does

Module S3 is the core of SPAN 2. It takes the 2,701 historical scenarios from S2 and replays each one against the portfolio, computing profit or loss for each scenario. It then finds the 99th percentile loss — the HVaR.

S3.1 Futures Scenario P&L

For each scenario date t and each futures position:

```
Position P&L(t) = dollar_move(t, product) × point_value × net_position
```

Example: On 2020-02-28 (COVID crash), CL moved -$7.579/bbl. CL_Mar25 (long 5): -$7.579 × $1,000 × 5 = -$37,895

Portfolio P&L is simply the sum across all positions on the same date. This automatically captures correlation — on Feb 28 2020, CL crashed AND GC crashed AND NG fell, so all long positions lost money simultaneously.

S3.2 The HVaR Calculation

```
HVaR(99%) = max(0, -P&L at 1st percentile of sorted portfolio P&L)
```

With 2,701 scenarios: 1% tail = 27 scenarios. HVaR = loss on the 27th worst day

```
Expected Shortfall (ES) = max(0, -average[P&L across worst 27 days])
```

ES = $50,392 in our model — the average loss if we land in the tail. Always ≥ HVaR.

Our results:

Metric	Value	Interpretation
Total Scenarios	2,701	Number of historical trading days used as scenarios
Tail Scenarios (1%)	27	The worst 27 days determine the HVaR
HVaR (99%)	$41,558	If today were one of our historical worst 1% days, the portfolio would lose $41,558
Expected Shortfall	$50,392	On average, the 27 worst days produce a $50,392 loss — $9,000 worse than the VaR cutoff
Worst Single Day	$-80,319	Feb 28 2020 COVID crash — worst day in 10yr history for this portfolio

S3.3 Tail Scenario Composition

Of the 27 worst scenarios: 16 are NORMAL market days (59.3%), 5 are ENERGY_SPIKE_2022 (18.5%), 3 are OIL_CRASH_2014 (11.1%), and 3 are COVID (11.1%).

> Key Insight: 59.3% of the worst 27 days are NORMAL (non-stress) market days. This means current elevated volatility — NG at 185% annualised — is generating losses as severe as the 2022 energy crisis. This is the vol-scaling mechanism working correctly: it elevates older calm-period returns to match today's high-vol environment.

S3.4 Long/Short Asymmetry

This is one of the most important outputs — a feature SPAN 1 cannot replicate:

Contract	Long HVaR	Short HVaR	Ratio (L/S)	Asymmetric?	Why?
CL_Mar25	$37,895	$37,232	1.018	No	CL returns are roughly symmetric (skewness = +0.076)
NG_Mar25	$26,266	$43,023	0.611	YES	NG has strong positive skew in price — 2022 spike means SHORT NG is riskier than long NG
GC_Apr25	$26,359	$27,572	0.956	No	GC mildly negative skew — nearly symmetric for this position size

The NG asymmetry ratio of 0.611 is striking: being short NG is 63% MORE expensive in margin than being long NG of the same size. This reflects the 2022 energy spike — natural gas prices tripling in 6 months created an enormous right tail (upside spike), which is catastrophic for short positions.

Part VI — Module S4: Stress VaR Engine

What This Module Does

Module S4 computes SVaR — the VaR within specifically defined crisis periods. It then blends HVaR and SVaR using the 40/60 weights to produce the primary risk number.

S4.1 SVaR Computation per Period

For each stress period, the module extracts portfolio P&L for dates within that window and computes VaR at 99% confidence within that sub-sample:

```
SVaR(period) = max(0, -P&L at 1% quantile within stress window)
```

Example: COVID window has 80 trading days. 1% of 80 = 0.8, rounded up to 1. SVaR = loss on the single worst COVID day.

Period	Days	SVaR	Worst Day	Worst P&L	Binding?
GFC	0	N/A	N/A	N/A	No — data starts 2015
OIL_CRASH_2014	157	$45,820	2015-12-30	$-45,820	No
COVID	80	$80,319	2020-02-28	$-80,319	YES
ENERGY_SPIKE_2022	147	$51,877	2022-05-09	$-51,877	No

S4.2 Why COVID is Binding

Feb 28 2020 decomposition:

Position	P&L on Feb 28 2020	Driver
CL_Mar25 (Long 5)	$-37,895	Oil crashed ~12% as global demand destruction became clear
CL_Jun25 (Short 3)	$+22,737	Short leg GAINS when oil crashes — partial offset
NG_Mar25 (Long 4)	$-16,969	Natural gas correlated selloff with oil
GC_Apr25 (Long 2)	$-48,193	CRITICAL — Gold also crashed on this day
PORTFOLIO TOTAL	$-80,319	All longs losing simultaneously — worst diversification breakdown

The GC story: Gold is supposed to be a safe haven. On Feb 28 2020 it crashed along with everything else. This is a liquidity crisis dynamic — when panic hits, investors sell EVERYTHING to raise cash, including gold. This is exactly the kind of correlation breakdown that only historical simulation (not fixed-scenario SPAN 1) can capture.

S4.3 HVaR / SVaR Blending

$$\text{Blended Raw} = (0.40 \times \text{HVaR}) + (0.60 \times \text{SVaR})$$

= (0.40 × $41,558) + (0.60 × $80,319) = $16,623 + $48,191 = $64,814

$$\text{Floor} = 0.80 \times \max(\text{HVaR}, \text{SVaR}) = 0.80 \times \$80{,}319 = \$64{,}255$$

Blended Floored = max($64,814, $64,255) = $64,815 (floor not binding)

The SVaR dominates at 74% of the blended total. This is intentional — in our portfolio, the COVID stress period represents a catastrophic outcome that the 10yr HVaR smooths out.

Part VII — Module S5: Volatility Surface Builder

What This Module Does

Module S5 builds an implied volatility surface for each product — a grid of implied volatilities across different strikes (moneyness) and expiry dates (tenors). This replaces the flat, hardcoded IVs used in SPAN 1.

S5.1 Why a Vol Surface?

In reality, options with different strikes do NOT all have the same implied volatility. OTM put options on crude oil are more expensive (higher IV) than ATM options because traders fear crashes more than rallies. This phenomenon is called the vol smile or vol skew.

Product	Skew Direction	Value (b)	Real-World Reason
CL (Crude Oil)	Negative skew	b = -0.08	Crude oil crashes faster than it rallies. Refiners buy put protection. OTM puts are expensive.
NG (Natural Gas)	Positive skew	b = +0.10	Natural gas can spike violently upward (winter shortages, geopolitical). OTM calls are expensive.
GC (Gold)	Mild negative skew	b = -0.03	Gold is a safe haven — mild put premium. Much flatter smile than energy.

S5.2 The Quadratic Smile Model

We fit a simple 3-parameter model to the vol surface:

$$\sigma(K) = a + b \times k + c \times k^2$$

Where k = ln(K/F) = log-moneyness, a = ATM vol level, b = skew (slope), c = curvature

Parameter	Name	Effect	CL 1M Value
a	Level	ATM implied volatility — the centre of the smile	54.42%
b	Skew	Slope: negative means puts more expensive than calls	b = -0.08
c	Curvature	How fast vol rises in the wings away from ATM	c = 0.60

S5.3 Vol Surface Output Explained

The surface table shows IV at 5 moneyness levels × 3 tenors for CL:

Moneyness	Strike (CL)	IV at 1M	IV at 3M	IV at 6M	Interpretation

K=80%	$63.36	59.19%	55.19%	49.19%	Deep OTM put — expensive due to crash risk
K=90%	$71.28	55.93%	51.93%	45.93%	OTM put — still elevated
K=100% (ATM)	$79.20	54.42%	50.42%	44.42%	Reference point — matches 60-day rolling vol
K=110%	$87.12	54.20%	50.20%	44.20%	OTM call — slightly cheaper (negative skew)
K=120%	$95.04	54.96%	50.96%	44.96%	Deep OTM call — smile curvature raises it back

Term Structure: Vol decreases as tenor increases (54.42% → 50.42% → 44.42% for CL ATM). This is called volatility backwardation — near-term uncertainty is higher than long-term. It is the normal state for commodity options. The term_slope parameter (-0.02 for CL) captures this.

S5.4 The Critical Impact: Flat IV vs Surface IV

Option	Flat IV (SPAN 1)	Surface IV (SPAN 2)	Difference	Premium Impact
CL_Mar25_C82	28.00%	54.21%	+2,621bp	$1,418 → $3,717 (+162%)
CL_Mar25_P76	26.00%	54.85%	+2,885bp	$1,051 → $3,429 (+226%)
GC_Apr25_C2100	18.00%	37.62%	+1,961bp	$2,287 → $7,518 (+229%)

The surface IV is approximately double the flat IV because the surface anchors to current realised volatility (56% for CL, 39% for GC) rather than stale hardcoded values. This is the most impactful practical difference in options margining.

Part VIII — Module S6: Options Engine

What This Module Does

Module S6 reprices all option positions under every historical scenario using the vol surface from S5, computes options HVaR, and combines it with futures HVaR to get a combined portfolio HVaR.

S6.1 Options Scenario P&L — The Full Reprice

For each of the 2,701 historical scenarios and each option:

4. Get the hypothetical futures price: F(t) = F(0) × exp(r_scaled(t))
5. Get the surface IV at the new moneyness K/F(t) using the interpolator
6. Reprice the option: Premium(t) = Black76(F(t), K, T, σ_surface(F(t), K, T))
7. P&L = (Premium(t) - Premium(0)) × point_value × net_position

This is fundamentally different from SPAN 1 which applies fixed ±5% vol shifts. Here, as the futures price moves, the option automatically moves to a different part of the vol surface — capturing delta, gamma, and vega effects simultaneously.

S6.2 The Diversification Benefit — The Key Result

Metric	Value	Interpretation
Futures HVaR (standalone)	$41,558	HVaR if we only had the futures positions
Options HVaR (standalone)	$19,009	HVaR if we only had the options positions
Sum of standalones	$60,567	What margin would be if futures and options were independent
Combined HVaR	$35,738	Actual portfolio HVaR — much lower due to hedging
Diversification Benefit	$24,829 (41%)	The options hedge the futures in the worst tail scenarios

> Why is there a 41% diversification benefit? On the worst futures days (big oil/gas crashes), the options portfolio PROFITS: the long put gains as oil falls, and the short calls gain as they go deeper OTM. The futures/options tail correlation is -0.893 — nearly perfectly offsetting in the tail. SPAN 2 captures this naturally through joint historical simulation. SPAN 1 does not.

S6.3 Options Attribution = 0% in Tail

In the 27 worst combined portfolio days, options P&L attribution = $0.00 for all three options. This means: on those 27 worst days, the options positions were collectively profitable (not contributing to losses). The pain was entirely in the futures book: NG_Mar25 (71.8%) and GC_Apr25 (51.4%).

Part IX — Module S7: Liquidity & Concentration Charges

What This Module Does

Module S7 adds two additive charges on top of the blended HVaR/SVaR risk measure. These charges cannot be offset by spreading credits — they represent real-world execution costs.

S7.1 Liquidity Charge

When you hold a large position and need to exit (e.g., margin call), you trade at worse-than-mid prices. The liquidity charge represents the cost of crossing the bid-ask spread:

```
Liq Charge = 0.5 × bid_ask_pct × notional × √(liquidity_horizon)
```

0.5 = we charge half the spread (one-way crossing cost). √horizon scales for multi-day unwind.

Notional calculation:

```
Notional = lots × current_price × point_value
```

CL_Mar25: 5 × $79.20 × 1,000 = $396,000. At 0.02% bid-ask: cost = $79.20. Charge = 50% × $79.20 = $39.60

Contract	Lots	Notional	Bid-Ask%	Liq Charge	Note
CL_Mar25	5	$396,000	0.020%	$39.60	CL is very liquid — tight spreads
CL_Jun25	3	$240,300	0.020%	$24.03	Deferred month — slightly less liquid but same spread rate
NG_Mar25	4	$108,400	0.050%	$27.10	NG has wider spreads due to volatility
GC_Apr25	2	$405,000	0.010%	$20.25	Gold very liquid — tightest spread
TOTAL	—	—	—	$110.98	Negligible — 0.17% of blended risk

S7.2 Concentration Charge

```
Threshold lots = ADV × concentration_threshold
```

CL: 350,000 × 0.5% = 1,750 lots. Our position = 5 lots. Not triggered.

```
Conc Charge = max(0, lots - threshold_lots) × HVaR_per_lot × conc_rate
```

Since 5 lots << 1,750 lot threshold, excess = 0 and charge = $0 for all positions

Scaling insight: At institutional scale (5,000 CL lots), the concentration threshold would be triggered: excess = 5,000 - 1,750 = 3,250 lots. Charge = 3,250 × $4,580/lot × 25% = $3.7 million additional margin. This is why SPAN 2 concentration charges matter enormously for large market participants but are zero for retail-sized portfolios like ours.

S7.3 Options Liquidity Charge

Options are quoted in volatility terms, so their liquidity cost uses vega (sensitivity to vol) rather than notional:

```
Opt Liq Charge = 0.5 × vega_per_lot × spread_vol × lots × point_value
```

GC_Apr25_C2100: 0.5 × 2.7751 × 0.010 × 1 × $100 = $1.39

Total options liq = $4.69. Tiny — because our option positions are small (1-3 lots) and the vega-spread product is small.

Part X — Module S8: Final Margin Aggregator & Comparison

The Complete SPAN 2 Margin Calculation

Module S8 assembles all components into the final SPAN 2 margin and produces the definitive side-by-side comparison against SPAN 1.

S8.1 Final Margin Formula Applied

Step	Component	Calculation	Value
1	Combined HVaR (futures + options)	From Module S6	$35,738
2	HVaR contribution	$35,738 × 40%	$14,295
3	Overall SVaR (COVID binding)	From Module S4	$80,319
4	SVaR contribution	$80,319 × 60%	$48,191
5	Blended Raw	$14,295 + $48,191	$64,487
6	Floor check	80% × max($35,738, $80,319) = 80% × $80,319	$64,255
7	Blended Floored	max($64,487, $64,255)	$64,815 (floor not binding)
8	Futures Liq Charge	From Module S7	$110.98
9	Concentration Charge	Not triggered	$0.00
10	Options Liq Charge	From Module S7	$4.69
11	SPAN 2 TOTAL MARGIN	$64,815 + $115.67	~$64,931

S8.2 The Full SPAN 1 vs SPAN 2 Comparison

Component	SPAN 1	SPAN 2	Key Difference
Gross risk (before credits)	$71,000	$60,567	SPAN 2 uses HVaR — real history gives a different starting point than fixed scan ranges
Spreading / Diversification credit	-$39,700	-$24,829	SPAN 1: fixed credit rates. SPAN 2: natural portfolio correlation from joint history

Net market risk (post credit)	$31,300	$35,738	SPAN 2 net is higher — less spreading credit but better tail capture
Stress component (SVaR)	$0	$48,191	SPAN 1 has NO stress component. SVaR is 74% of SPAN 2 final margin.
Blended risk	$31,300	$64,815	The stress component more than doubles the margin
Liquidity + Concentration	$0	$116	Negligible for retail size
TOTAL MARGIN	$52,366	~$64,931	SPAN 2 is 24% higher

S8.3 SPAN 2 Margin Composition

Component	Amount	% of Total	Insight
HVaR contribution (40%)	~$14,295	~22%	Current-market-calibrated risk — moderate
SVaR contribution (60%)	~$48,191	~74%	COVID stress dominates — the binding constraint
Liquidity + Concentration	$116	~0.2%	Negligible at retail scale — massive at institutional scale

Part XI — How Close Are We to Real CME SPAN 2?

Honest Assessment: What We Replicate and What We Approximate

This is the most important section for making informed decisions about the model's outputs. Our SPAN 2 implementation replicates the conceptual architecture of the real CME system but makes several practical simplifications.

What We Replicate Correctly (High Fidelity)

Feature	Our Implementation	Fidelity	Notes
HVaR framework	99% confidence, 10yr history, vol-scaling	High	Core methodology matches CME SPAN 2 design
SVaR concept	Worst VaR across defined stress periods	High	Correct concept; our stress periods are well-chosen
HVaR/SVaR blending	40%/60% weighted blend with 80% floor	High	Parameters are within CME's disclosed range
Long/short asymmetry	Naturally from signed P&L distribution	High	This is exactly how SPAN 2 captures it — no approximation
Vol surface shape	Quadratic smile with product-specific skew	Medium-High	Shape is correct; calibration is synthetic
Options full reprice	Black-76 with surface vol per scenario	High	Correct methodology
Diversification from history	Natural from joint scenario replay	High	This is precisely the SPAN 2 approach
Liquidity charge concept	Bid-ask × notional × multiplier	Medium	Concept correct; real CME has more granular calibration
Concentration charge concept	ADV threshold, excess × per-lot risk	Medium	Concept correct; real CME uses more sophisticated market impact models

Where We Simplify (Lower Fidelity)

Feature	Real CME SPAN 2	Our Model	Impact of Simplification
Price data source	Full futures curve (all expiries) with proper roll	Front-month yfinance ticker (auto-roll)	May understate basis risk between expiries. Small impact for front-month-dominant portfolios.

Scenario count	CME uses full available history (potentially 15+ years)	10yr yfinance data (2,763 days)	Slightly fewer extreme scenarios. GFC (2008) missing entirely.
GFC stress period	Available in CME's dataset	Missing — data starts 2015	SVaR binding period may change. COVID likely still binding.
Vol surface calibration	Live options market data, 20+ strikes per expiry	Synthetic surface from empirical skew params	Surface shape is directionally correct but not market-fitted. IV levels driven by realised vol, not market IV.
Options SVaR	Options fully repriced under stress scenarios	Not implemented — futures SVaR only	Understates options stress risk marginally
Cross-margining	Offsets across CME + CBOT + clearing members	Single portfolio only	No cross-margin benefit modelled
Multi-currency	FX conversion for non-USD products	All USD	Not relevant for our product set
Intraday margining	CME runs SPAN 2 on intraday price moves too	End-of-day only	Intraday margin calls not captured
Parameter file format	CME publishes daily .pa2 files with exact parameters	Hardcoded/estimated parameters	Our parameters are plausible but not CME-official
Concentration model	Full market impact model (Almgren-Chriss)	Simple linear threshold	Significantly underestimates concentration cost for large positions

What The Numbers Tell Us About Fidelity

Metric	Our Model	Likely Real CME Range	Assessment
HVaR confidence	99%	99%	Exact match
HVaR lookback	10yr	10yr minimum	Match
SVaR binding period	COVID ($80,319)	Likely COVID or similar	Plausible — COVID was globally binding
HVaR/SVaR blend	40%/60%	CME has not disclosed exact weights	Within reasonable range
Total margin vs SPAN 1	1.24× ($64,931 vs $52,366)	Real SPAN 2 migration showed -10% to +20% depending on portfolio	Our result is within the observed range

NG long/short asymmetry	0.611 ratio	Real NG is strongly asymmetric — qualitative match	Directionally correct
Options premium (surface vs flat)	~2× uplift	Real uplift varies — surface is dynamic	Likely overstated since our surface uses realised vol, not market IV

Part XII — Reading Every Output Line

Field-by-Field Reference for All 8 Module Outputs

Module S2 Output Fields

Field	What It Shows	Example	How to Use It
Observations	Number of trading days downloaded	2,763	More is better. Fewer than 1,000 would be concerning for HVaR reliability.
First/Last Date	Date range of price history	2015-03-23 → 2026-03-18	Confirms the 10yr window. Note GFC (2008) is not included.
Current Volatility	60-day rolling annualised vol	CL: 56.4%, NG: 185.2%	High vol → larger scenario moves → higher HVaR. NG at 185% is extreme.
Total Scenarios	Post-dropna scenario count	2,701	This is the 'n' in your HVaR calculation. 1% tail = 27 scenarios.
Stressed Scenarios	Days tagged per stress period	COVID: 80 days	These will be extracted for SVaR in S4.
Skewness	Directional asymmetry of returns	NG: -1.005	Negative = long riskier. Positive = short riskier. |>0.5| = significant asymmetry.
Excess Kurtosis	Fat-tail measure	CL: 22.262	Higher = more extreme events. Normal = 0. Values of 20+ indicate very fat tails.
Worst/Best 1% Return	Tail return quantiles	CL worst 1%: -7.36%	These translate directly to HVaR: -7.36% × $79.20 × $1,000 × 5 = $29,150.

Module S3 Output Fields

Field	What It Shows	Our Value	Interpretation
HVaR (99%)	Primary margin component	$41,558	The minimum margin needed to cover 99% of historical scenarios for futures only.
Expected Shortfall	Average loss in tail	$50,392	If margin is breached, expected shortfall is $50,392. The $8,834 gap above HVaR measures tail severity.
Tail Scenario Composition	Which periods drive worst 27 days	59.3% Normal	High normal% means current vol is as dangerous as named crises.
Position Attribution	Each position's share of HVaR	NG: 71.8%, GC: 51.4%	Attribution > 100% sum is possible when some positions are negative (offsetting). Identifies dominant risk drivers.
Long HVaR / Short HVaR	Asymmetric margin by direction	NG ratio = 0.611	Ratio < 1 means short is riskier. Ratio > 1 means long is riskier. Used in SPAN 2 to differentiate margin by direction.
P&L Histogram	Distribution shape	Skewness = 0.296	Positive portfolio skew despite negative product skews — short CL_Jun25 inverts its contribution.

Module S4 Output Fields

Field	What It Shows	Our Value	Interpretation

SVaR by period	VaR within each stress window	COVID: $80,319	Tells you which crisis was worst for THIS specific portfolio.
Binding Period	The crisis that dominates	COVID	SVaR = $80,319. This single number drives 74% of final SPAN 2 margin.
Worst Day	Worst calendar date in binding period	2020-02-28	Useful for post-mortem: what happened on that day and why?
SVaR Attribution	Position attribution in stress tail	GC: 60% of SVaR	Gold dominated on Feb 28 2020 because it crashed alongside oil — correlation breakdown.
Blended Risk	Final pre-charge risk number	$64,815	= 40%×$35,738 + 60%×$80,319. This is the starting point for S8.
Floor Binding?	Whether 80% floor applied	No	Floor = $64,255 < Blended = $64,815. Floor not needed here.

Module S5 Output Fields

Field	What It Shows	Interpretation
Vol Surface Grid	IV at each (moneyness, tenor) cell	Read across a row to see the smile shape. Read down a column to see term structure.
ATM Vol	Implied vol at K=100%	Anchored to 60-day rolling realised vol. CL: 54.42%, not the 28% hardcoded in SPAN 1.
Skew (b)	Smile slope parameter	Negative b = left skew (puts expensive). Positive b = right skew (calls expensive). Size indicates steepness.
Curvature (c)	Smile convexity parameter	Higher c = deeper smile. Wings are more expensive relative to ATM.
RMSE	Fit quality of quadratic model	0.0000% — perfect fit with 3 params to 5 points. Would be ~1% with real market data.

ASCII Smile	Visual representation of smile shape	Read top-to-bottom as OTM-put to OTM-call. Asymmetry shows skew direction.

Module S6 Output Fields

Field	What It Shows	Our Values	Interpretation
Flat IV vs Surface IV	SPAN 1 vs SPAN 2 vol for each option	Diff: +2,600bp for CL	Surface IV is ~2× flat IV because surface anchors to current realised vol.
IV Diff (bp)	Basis point difference in vol	CL C82: +2,621bp	Directly translates to premium uplift. 2,600bp on a $1,418 option ≈ +$2,300 premium.
Delta (Surface)	Black-76 delta using surface vol	CL C82: 0.4420	Higher surface vol → higher delta for OTM options. Option is more sensitive to price.
Prem (Surface) vs Prem (Flat)	Dollar premium under each vol model	$3,717 vs $1,418	The surface premium is the correct current market value. Flat IV understates risk.
Futures/Options Correlation in Tail	-0.893	Tail correlation between futures P&L and options P&L	Strong negative: options profit when futures lose. The portfolio hedge is working.
Diversification Benefit	$24,829 (41%)	Margin reduction from joint simulation	41% is large — driven by the -0.893 tail correlation. SPAN 1 misses this entirely.

Module S7 Output Fields

Field	What It Shows	Interpretation
Notional	Position value in dollars	Lots × price × point_value/price × point_value. Used as base for liq charge.
Bid-Ask%	Bid-ask spread as % of price	CL: 0.02%, NG: 0.05%, GC: 0.01%. Wider spreads = higher liq charge.
B-A Cost	Total bid-ask cost to unwind	Notional × bid-ask%. Represents cost if you traded through the full spread.
Liq Charge	Final liquidity charge per position	50% × B-A Cost × √horizon. We charge half the spread (one-way cost).
Threshold Lots	ADV × concentration threshold	CL: 1,750 lots. Our position (5 lots) is 0.3% of threshold.

HVaR/Lot	Per-lot HVaR attribution	CL: $4,580/lot, GC: $9,330/lot. GC has highest per-lot risk despite lowest total risk.
Active?	Whether concentration charge triggered	No for all — retail portfolio too small.
Vega/Lot (options)	Vega per option lot from surface greeks	GC call: 2.7751. Means $277.51 P&L per 1 vol point per lot.

Module S8 Output Fields

Field	What It Shows	Interpretation
Combined HVaR	HVaR of full portfolio (futures+options)	$35,738. Lower than futures-only ($41,558) because options provide a hedge.
Overall SVaR	SVaR from binding stress period	$80,319. COVID Feb 28 2020 drives this number.
Blended Raw	Weighted average before floor	40%×$35,738 + 60%×$80,319 = $64,815.
Floor	80% × max(HVaR, SVaR)	$64,255. Not binding since blended raw ($64,815) > floor.
Total Margin	Final SPAN 2 requirement	~$64,931. Compare to SPAN 1's $52,366 — SPAN 2 is 24% higher.
Ratio SPAN2/SPAN1	Margin multiple	~1.24×. Within the observed real-world migration range of 0.9× to 1.2×.
SVaR as % of total	How dominant the stress component is	~74%. The COVID stress period is the primary margin driver for this portfolio.

— End of SPAN 2 Guidebook —

8 Modules · Real Market Data · 2,701 Historical Scenarios · March 2026